AI 101 for Mechanical Engineers · Module 10

Leakage is information crossing the wrong boundary

Lesson 10.3

What you will learn

> Identify preprocessing, temporal, target-derived and duplicate-event leakage, explain how training-fold pipelines protect evaluation, and name the records needed to reproduce the procedure.

Lesson overview

Module 10 · Lesson 3

Leakage is information crossing the wrong boundary

AI 101 for Mechanical Engineers

1 Check four leakage routes

1 Preprocessing
A scaler or imputer fitted on all data lets held-out examples shape training.

2 Future readings
A centred window at time *t* can include the reading at *t*+1.

3 Outcome-derived inputs
A post-failure repair code is unavailable for an earlier warning.

4 Duplicate events
Related events in development and evaluation share information.

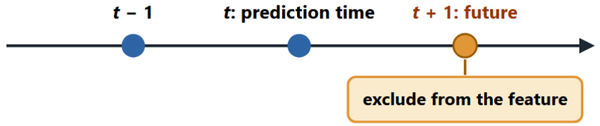
A surprisingly strong result is a reason to audit, not proof of leakage.

2 Fit only inside training folds

1	2	3	4
Form the folds	Fit transforms on the training fold	Store the fitted parameters	Apply to held-out examples without refitting

Calling it a pipeline is not enough: check which examples fitted each transformation.

Check availability at prediction time



A chronological split alone does not repair a feature that already contains future data.

3 Test and report honestly

Complete development → Lock the procedure → Evaluate the reserved test

If test results guide further changes, disclose the reuse and seek new independent evidence.

Record the experiment

- ✓ Split identifiers
- ✓ Transformation sequence
- ✓ Random seeds
- ✓ Code versions
- ✓ Software dependencies

Repeating a score does not prove the boundary was valid.

Key takeaway — **Inspect both dataset separation and the information available when the prediction is made.**

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Leakage is information crossing the wrong boundary

- Unavailable or reserved information improperly influences training or selection [25], [58]
- Boundaries can concern datasets, machines, events or prediction time
- It concerns information and timing, not dataset filenames

Fit preprocessing after forming the folds

Incorrect

- Fit scaler or imputer on all data
- Then form the evaluation folds
- Held-out folds influence the fitted values

Correct

- Form the folds first
- Fit on the training fold
- Apply those fitted values to evaluation

> Preprocessing can leak without using target labels.

A pipeline must keep fitting inside training

- Groups transformations and model processing into a repeatable sequence
- Fitted on training; stored parameters applied to evaluation examples
- Refit separately within each cross-validation training fold
- Calling it a pipeline is insufficient if fitted on all examples

A centered average can use future readings



- A live alarm at t must use information available then
- The reading at t+1 is unavailable when the alarm operates
- A chronological split alone does not repair this feature

Four leakage routes need different inspections

Route	Inspect
Preprocessing	Which data fitted the transformation
Temporal	Availability at prediction time
Target-derived	Outcome information, e.g. post-failure repair code
Duplicate event	Separation of related examples and windows

> Check event identity and window overlap, not only individual row assignments.

A surprising result is a reason to audit

- Unexpectedly strong or sharply changed results: audit, not proof
- Trace each feature: would it exist at prediction time?
- Learn target encoding only within the relevant training fold
- After deployment, shift can degrade performance without leakage [59]

Test once, after development is complete

1. Complete development
2. Fix the procedure
3. Evaluate the reserved final test
4. Report the procedure actually executed
5. Disclose any test reuse; seek new independent evidence where possible

Record enough to reproduce the procedure

Record	Identifies
Split identifiers	Which examples were in each portion
Transformation sequence	The processing steps
Random seeds	Random initialization or selection used
Code and dependency versions	Implementation and environment

> Records let the procedure be repeated and inspected [36]; a reproduced score alone does not prove no leakage.

CHECK YOUR UNDERSTANDING

Why might a centered moving-average feature leak in a real-time alarm?

Answer before moving on.

CHECK YOUR UNDERSTANDING

Why might a centered moving-average feature leak in a real-time alarm?

> It can include a reading after the alarm’s prediction time. That information is unavailable when the alarm operates, even with a chronological split.

CHECK YOUR UNDERSTANDING

Why must a scaler be fitted again inside each cross-validation training fold?

Answer before moving on.

CHECK YOUR UNDERSTANDING

Why must a scaler be fitted again inside each cross-validation training fold?

> Each iteration has a different held-out portion. Fitting only on that iteration’s training examples stops the held-out portion influencing the transformation.

Key takeaways

- Inspect dataset separation and information availability at prediction time
- Fit preprocessing in training folds; apply stored parameters to evaluation
- Exclude unavailable future and outcome-derived information from inputs
- Disclose final-test reuse; document enough for reproduction

References

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[36] J. Pineau et al., “Improving reproducibility in machine learning research (A report from the NeurIPS 2019 reproducibility program),” J. Mach. Learn. Res., vol. 22, no. 164, pp. 1–20, 2021.

[58] S. Kapoor and A. Narayanan, “Leakage and the reproducibility crisis in machine-learning-based science,” Patterns, vol. 4, no. 9, Art. no. 100804, 2023. doi: 10.1016/j.patter.2023.100804

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Numbers match the course reference list. The written lesson is the authoritative text.